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Inventor(s)	Watanabe; Takaharu et al.

Fuel cell vehicle and prediction method

Abstract

A fuel cell vehicle includes a first current amount calculating unit that calculates a first total amount of current output from a fuel cell during a period from a predetermined timing to the present time, a hydrogen amount calculating unit that calculates a hydrogen total amount output from a hydrogen tank during the period from the predetermined timing to the present time, a remaining amount acquisition unit that acquires a hydrogen remaining amount in the hydrogen tank, and a current amount prediction unit that predicts a current amount that can be supplied by the fuel cell based on the first total amount, the total amount, and the hydrogen remaining amount.

Inventors: **Watanabe; Takaharu (Tokyo, JP), Suzuki; Kenta (Tokyo, JP), Mochizuki; Hideki (Tokyo, JP), Yamanaka; Suguru (Tokyo, JP), Fukunishi; Yuki (Tokyo, JP)**

Applicant: **HONDA MOTOR CO., LTD.** (Tokyo, JP)

Family ID: **1000008779871**

Assignee: **Honda Motor Co. Ltd. (Tokyo, JP)**

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Primary Examiner: Lee; Tyler J

Attorney, Agent or Firm: Carrier, Shanda & Associates P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from Japanese Patent

Application No. 2022-052680 filed on Mar. 29, 2022, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a fuel cell vehicle and a prediction method for predicting an amount of electric current that can be supplied by a fuel cell.

Description of the Related Art

(3) In recent years, research and development have been conducted on fuel cells that contribute to energy efficiency in order to ensure that more people have access to affordable, reliable, sustainable and modern energy.

(4) JP 2021-118658 A discloses a method for calculating a cruising distance of a fuel cell vehicle including a fuel cell. According to this method, the cruising distance of the fuel cell vehicle is calculated based on the fuel efficiency of the fuel cell vehicle.

SUMMARY OF THE INVENTION

(5) In the technique related to the fuel cell, it may be difficult to accurately calculate the cruising distance based on the fuel consumption.

(6) An object of the present invention is to solve the aforementioned problem.

(7) According to a first aspect of the present invention, a fuel cell vehicle is equipped with a hydrogen tank configured to store hydrogen, a fuel cell configured to generate electric power using the hydrogen, and a drive source configured to be driven using electric power generated by the fuel cell, and the fuel cell vehicle includes a first current amount calculating unit configured to calculate a first total amount of output current that is output by the fuel cell during a period from a predetermined timing to present time, a hydrogen amount calculating unit configured to calculate a total amount of hydrogen that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time, a remaining amount acquisition unit configured to acquire a remaining amount of hydrogen remaining in the hydrogen tank at the present time, and a current amount prediction unit configured to predict a current amount that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen.

(8) According to a second aspect of the invention, a prediction method predicts an amount of current that is configured to be supplied by a fuel cell of a fuel cell vehicle, the fuel cell vehicle including a hydrogen tank configured to store hydrogen, the fuel cell configured to generate electric power using the hydrogen, and a drive source configured to be driven using electric power generated by the fuel cell. The prediction method includes a first current amount calculating step of calculating a first total amount of output current that is output by the fuel cell during a period from a predetermined timing to present time, a hydrogen amount calculating step of calculating a total amount of hydrogen that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time, a remaining amount acquiring step of acquiring a remaining amount of hydrogen remaining in the hydrogen tank at the present time, and a current amount predicting step of predicting a current amount that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen.

(9) According to the present invention, since it is possible to accurately predict the amount of current that the fuel cell of the fuel cell vehicle can supply to the drive source, it is possible to accurately calculate the cruising distance based on the amount of current.

(10) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a configuration diagram of a vehicle according to an embodiment;
(2) FIG. **2** is a configuration diagram of a prediction device; and
(3) FIG. **3** is a flowchart illustrating a flow of a prediction method according to the embodiment.
DETAILED DESCRIPTION OF THE INVENTION
Embodiment
(4) FIG. **1** is a configuration diagram of a vehicle **10** according to the embodiment.
(5) The vehicle **10** is a fuel cell vehicle that includes a hydrogen tank **12**, a fuel cell **14**, a battery **16**, a drive source **18**, a display unit **20**, and a prediction device **22**.
(6) The hydrogen tank **12** is a tank that stores hydrogen (hydrogen gas). The hydrogen stored in the hydrogen tank **12** is supplied to the fuel cell **14**.
(7) The fuel cell **14** is a power generation device that generates power using hydrogen supplied from the hydrogen tank **12**. The fuel cell **14** generates electric power using a chemical reaction between the supplied hydrogen and oxygen. For example, air containing oxygen is supplied to the fuel cell **14**. The fuel cell **14** outputs electric current by generating electric power. The output current of the fuel cell **14** is input to the drive source **18** or the battery **16**.
(8) The battery **16** is a secondary battery that stores electric power. The battery **16** is charged when a current is supplied thereto. The charged battery **16** can supply current to the drive source **18** as needed.
(9) The drive source **18** is, for example, a rotating electric machine. The drive source **18** is driven using the supplied current. The drive source **18** is driven to rotate wheels of the vehicle **10** via, for example, a predetermined transmission mechanism. In accordance therewith, the vehicle **10** travels. That is, the vehicle **10** travels in accordance with driving of the drive source **18**. The predetermined transmission mechanism includes, for example, a transmission. The wheels and the predetermined transmission mechanism are not shown in the accompanying drawings.
(10) When the vehicle **10** is decelerated, the rotating electric machine included in the drive source **18** may generate electric power using rotational force (regenerative force) of the wheels. The current generated by the power generation of the rotating electric machine may be supplied to the battery **16**.
(11) The display unit **20** is, for example, a multi information display (MID). The display unit **20** includes a display screen **201** that appropriately displays various types of information. The display unit **20** may be a liquid crystal display, an organic EL display, or the like.
(12) The vehicle **10** may further include an operating unit that is operated by a vehicle occupant to switch the type of information to be displayed on the display screen **201**.
(13) FIG. **2** is a configuration diagram of the prediction device **22**.
(14) The prediction device **22** is a computer that predicts an integrated current amount that can be generated by the fuel cell **14**. The prediction device **22** is included in, for example, an in-vehicle electronic control device (ECU). The prediction device **22** includes a storage unit **24** and a computation unit **26**.
(15) The storage unit **24** includes a storage circuit. The storage circuit includes, for example, one or more memories such as a random access memory (RAM) and a read only memory (ROM).
(16) The storage unit **24** stores a prediction program **28**. The prediction program **28** is a program for causing the prediction device **22** to implement the prediction method according to the present embodiment.
(17) The data stored in the storage unit **24** is not limited to the prediction program **28**. The storage unit **24** may store various types of data as necessary. Some of the various data stored in the storage unit **24** will be described later.

(18) The computation unit **26** includes processing circuitry. The processing circuitry includes, for example, one or more processors. However, the processing circuitry of the computation unit **26** may include an integrated circuit such as an application specific integrated circuit (ASIC) or a field-programmable gate array (FPGA). The processing circuitry of the computation unit **26** may include a discrete device.
(19) The computation unit **26** includes a first current amount calculating unit **30**, a hydrogen amount calculating unit **32**, a remaining amount acquisition unit **34**, a current amount prediction unit **36**, a travel distance calculating unit **38**, a second current amount calculating unit **40**, an electric power consumption calculating unit **42**, a remaining capacity acquisition unit **44**, a cruising distance calculating unit **46**, and a display control unit **48**.
(20) The first current amount calculating unit **30**, the hydrogen amount calculating unit **32**, the remaining amount acquisition unit **34**, the current amount prediction unit **36**, and the travel distance calculating unit **38** are realized by the processor of the computation unit **26** executing the prediction program **28**. Similarly, the second current amount calculating unit **40**, the electric power consumption calculating unit **42**, the remaining capacity acquisition unit **44**, the possible distance calculating unit **46**, and the display control unit **48** are realized by the processor of the computation unit **26** executing the prediction program **28**. However, the integrated circuit, the discrete device, or the like described above may realize at least a part of the first current amount calculating unit **30**, the hydrogen amount calculating unit **32**, the remaining amount acquisition unit **34**, the current amount prediction unit **36**, and the travel distance calculating unit **38**. The integrated circuit, the discrete device, or the like described above may realize at least a part of the second current amount calculating unit **40**, the electric power consumption calculating unit **42**, the remaining capacity acquisition unit **44**, the cruising distance calculating unit **46**, and the display control unit **48**.
(21) The first current amount calculating unit **30** calculates a total amount of output current that is output by the fuel cell **14** during a predetermined period. The total amount of the output current that is output by the fuel cell **14** during the predetermined period is also referred to as a first total amount CA**1** in the following description. The calculated first total amount CA**1** may be memorized in the storage unit **24**.
(22) The predetermined period is, for example, a period from a predetermined timing to the present time. The predetermined timing is, for example, the time when the vehicle **10** was started. However, the predetermined timing may be set to time while the vehicle **10** is traveling.
(23) The first total amount CA**1** is calculated based on, for example, a detection signal output by a current sensor **50** (see FIG. **1**) in accordance with the output current that is output by the fuel cell **14**. The first current amount calculating unit **30** calculates the first total amount CA**1** by integrating the current detected by the current sensor **50**.
(24) The hydrogen amount calculating unit **32** calculates a total amount CHG of hydrogen (hydrogen total amount CHG) output from the hydrogen tank **12** to the fuel cell **14** during the predetermined period. The calculated hydrogen total amount CHG may be memorized in the storage unit **24**.
(25) The hydrogen total amount CHG is calculated based on, for example, a detection signal output by a flow rate sensor **52** (see FIG. **1**) in accordance with the flow rate of hydrogen output from the hydrogen tank **12**. The hydrogen amount calculating unit **32** calculates the hydrogen total amount CHG by integrating the flow rate detected by the flow rate sensor **52**.
(26) The hydrogen total amount CHG may be acquired using a pressure sensor that detects the gas pressure in the hydrogen tank **12**. That is, when hydrogen is output from the hydrogen tank **12**, the gas pressure in the hydrogen tank **12** changes. The flow rate of hydrogen can be calculated based on the amount of change in the gas pressure.
(27) The remaining amount acquisition unit **34** acquires a remaining amount RHG of hydrogen (hydrogen remaining amount RHG) currently remaining in the hydrogen tank **12**. The acquired hydrogen remaining amount RHG may be memorized in the storage unit **24**.

(28) The hydrogen remaining amount RHG can be acquired based on the detection signal of the flow rate sensor **52** or the pressure sensor, similarly to the hydrogen total amount CHG.
(29) The current amount prediction unit **36** predicts the amount of current that can be supplied by the fuel cell **14** based on the first total amount CA**1**, the hydrogen total amount CHG, and the hydrogen remaining amount RHG. The current amount predicted by the current amount prediction unit **36** is also referred to as a predicted current amount CAP in the following description.
(30) The predicted current amount CAP is a product of a quotient obtained by dividing the first total amount CA**1** by the hydrogen total amount CHG and the hydrogen remaining amount RHG. The current amount prediction unit **36** predicts the predicted current amount CAP based on, for example, the following equation (1). In Equation (1), CAP indicates the predicted current amount CAP (unit: ampere hour). CA**1** indicates the first total amount CA**1** (unit: ampere hour) during the predetermined period. CHG indicates the hydrogen total amount CHG (unit: gram) during the predetermined period. RHG indicates the current hydrogen remaining amount RHG (unit: gram).
(31) $\mathrm{CAP} = \frac{\mathrm{CA1}}{\mathrm{CHG}} \times \mathrm{RHG}$ (1)
(32) The calculated predicted current amount CAP may be stored in the storage unit **24**.
(33) According to the current amount prediction unit **36**, the predicted current amount CAP is accurately predicted. The predicted current amount CAP can be used to calculate a cruising distance (distance to empty) CD of the vehicle **10**, as described further below.
(34) The travel distance calculating unit **38** calculates a travel distance TD. The travel distance TD is a distance by which the vehicle **10** has traveled during the predetermined period. The calculated travel distance TD may be memorized in the storage unit **24**.
(35) The travel distance calculating unit **38** calculates, for example, a product of a vehicle speed (velocity) and the predetermined period (time) as the travel distance TD. The vehicle speed is acquired based on, for example, a detection signal output by a vehicle speed sensor **54** (see FIG. **1**) in response to the output of the drive source **18**.
(36) The second current amount calculating unit **40** calculates a total amount of output current that is output by the battery **16** during the predetermined period. The total amount of the current that is output by the battery **16** during the predetermined period is also referred to as a second total amount CA**2** in the following description. The calculated second total amount CA**2** may be memorized in the storage unit **24**.
(37) The output current of the battery **16** is acquired based on, for example, a detection signal that is output by a current sensor **56** (see FIG. **1**) in accordance with the output current of the battery **16**. The second current amount calculating unit **40** calculates the second total amount CA**2** by integrating the current detected by the current sensor **56**.
(38) The electric power consumption calculating unit **42** calculates an electric power consumption EPC of the vehicle **10** based on the first total amount CA**1**, the second total amount CA**2**, and the travel distance TD.
(39) The electric power consumption EPC is a quotient obtained by dividing the travel distance TD by the sum of the first total amount CA**1** and the second total amount CA**2**. The electric power consumption calculating unit **42** calculates the electric power consumption EPC based on, for example, the following equation (2). In Equation (2), EPC indicates the electric power consumption EPC (unit: meter/ampere hour). CA**2** indicates the second total amount CA**2** (unit: ampere hour). TD indicates the travel distance TD (unit: meter) of the vehicle **10** during the predetermined period.
(40) $\mathrm{EPC} = \frac{\mathrm{TD}}{\mathrm{CA1} + \mathrm{CA2}}$ (2)
(41) According to Equation (2), the electric power consumption EPC indicates the travel distance of the vehicle **10** per unit current amount. The calculated electric power consumption EPC may be memorized in the storage unit **24**.
(42) The remaining capacity acquisition unit **44** acquires the current remaining capacity CA**3** of the battery **16**. The remaining capacity CA**3** is acquired based on, for example, a detection signal

output by a remaining capacity sensor **58** (see FIG. **1**) in accordance with the voltage and the current of the battery **16**. The acquired remaining capacity CA**3** may be memorized in the storage unit **24**.

(43) The cruising distance calculating unit **46** calculates the cruising distance CD of the vehicle **10** based on the predicted current amount CAP, the electric power consumption EPC, and the remaining capacity CA**3**. The cruising distance CD indicates a distance that the vehicle **10** can travel when both the fuel cell **14** and the battery **16** are used. The calculated cruising distance CD may be memorized in the storage unit **24**.

(44) The cruising distance CD is a sum of the product of the electric power consumption EPC and the remaining capacity CA**3** and the product of the electric power consumption EPC and the predicted current amount CAP. Therefore, the cruising distance calculating unit **46** calculates the cruising distance CD based on, for example, the following equation (3). In Equation (3), CD indicates the cruising distance CD (unit: meter). CA**3** indicates the current remaining capacity CA**3** (unit: ampere hour) of the battery **16**.

$$CD=EPC\times CA3+EPC\times CAP \qquad (3)$$

(45) According to Equation (3), the cruising distance CD is obtained based on the amount of current (the first total amount CA**1** and the second total amount CA**2**) and the electric power consumption EPC. Therefore, according to the present embodiment, the cruising distance CD of the vehicle **10** can be calculated without calculating the fuel consumption of the vehicle **10**.

(46) Incidentally, there is a method of calculating the cruising distance CD of the vehicle **10** based on fuel consumption. That is, the fuel consumption is a travel distance of the vehicle **10** per unit weight of hydrogen. Conventionally, the cruising distance CD is calculated based on the product of the fuel consumption and the hydrogen remaining amount RHG in the hydrogen tank **12** (see also JP 2021-118658 A).

(47) However, the fuel consumption changes while the vehicle **10** is traveling. For example, the above-described rotating electric machine generates electric power using regenerative power. Thus, the battery **16** is charged (referred to as Event 1). When the vehicle **10** travels using the charged battery **16**, the power generation amount of the fuel cell **14** decreases. As a result, the fuel efficiency is improved (referred to as Event 2) after Event 1.

(48) Here, a time difference between Events 1 and 2 causes a problem. In other words, there is a time difference between when the battery **16** is charged and when the power generation amount of the fuel cell **14** decreases. Therefore, at the stage of Event 1, it is impossible to calculate the cruising distance CD using the fuel consumption after Event 2. As a result, between Events 1 and 2, the cruising distance CD calculated based on the fuel consumption lacks accuracy.

(49) In this regard, according to the present embodiment, respective changes in the first total amount CA**1** of the fuel cell **14** and the remaining capacity CA**3** can be quickly reflected in the cruising distance CD. Thus, the accuracy of the calculated cruising distance CD can be maintained.

(50) The display control unit **48** controls the display unit **20** to display the cruising distance CD on the display screen **201**. Thus, the vehicle occupant of the vehicle **10** can learn the cruising distance CD. In addition, the display control unit **48** may control the display unit **20** to further display the predicted current amount CAP, the electric power consumption EPC, and the like on the display screen **201**.

(51) The display control unit **48** may switch the display of the predicted current amount CAP, the electric power consumption EPC, and the like in accordance with an instruction from the vehicle occupant.

(52) FIG. **3** is a flowchart illustrating a flow of a prediction method according to an embodiment.

(53) The prediction device **22** is capable of performing the prediction method of FIG. **3**. The prediction method of FIG. **3** includes a first current amount calculating step S**1**, a hydrogen amount calculating step S**2**, a remaining amount acquiring step S**3**, and a travel distance calculating step S**4**. The prediction method of FIG. **3** further includes a second current amount calculating step S**5**,

a remaining capacity acquiring step S**6**, a current amount predicting step S**7**, an electric power consumption calculating step S**8**, a cruising distance calculating step S**9**, and a display control step S**10**.

(54) However, the first current amount calculating step S**1**, the hydrogen amount calculating step S**2**, the remaining amount acquiring step S**3**, the travel distance calculating step S**4**, the second current amount calculating step S**5**, and the remaining capacity acquiring step S**6** may be performed in any order.

(55) In the first current amount calculating step S**1**, the first current amount calculating unit **30** calculates the first total amount CA**1**. The first current amount calculating unit **30** can calculate the first total amount CA**1** based on the detection signal of the current sensor **50**.

(56) In the hydrogen amount calculating step S**2**, the hydrogen amount calculating unit **32** calculates the hydrogen total amount CHG. The hydrogen amount calculating unit **32** can calculate the hydrogen total amount CHG based on the detection signal of the flow rate sensor **52**. However, a pressure sensor may be used instead of the flow rate sensor **52**.

(57) In the remaining amount acquiring step S**3**, the remaining amount acquisition unit **34** acquires the hydrogen remaining amount RHG. The remaining amount acquisition unit **34** can acquire the hydrogen remaining amount RHG based on the detection signal of the flow rate sensor **52** or the pressure sensor.

(58) In the travel distance calculating step S**4**, the travel distance calculating unit **38** calculates the travel distance TD. The travel distance calculating unit **38** can acquire the travel distance TD based on the detection signal of the vehicle speed sensor **54**.

(59) In the second current amount calculating step S**5**, the second current amount calculating unit **40** calculates the second total amount CA**2**. The second current amount calculating unit **40** can calculate the second total amount CA**2** based on the detection signal of the current sensor **56**.

(60) In the remaining capacity acquiring step S**6**, the remaining capacity acquisition unit **44** acquires the remaining capacity CA**3**. The remaining capacity acquisition unit **44** can acquire the remaining capacity CA**3** based on the detection signal of the remaining capacity sensor **58**.

(61) In the current amount predicting step S**7**, the current amount prediction unit **36** predicts the predicted current amount CAP based on the execution results of the first current amount calculating step S**1**, the hydrogen amount calculating step S**2**, and the remaining amount acquiring step S**3**. The current amount prediction unit **36** can predict the predicted current amount CAP by using, for example, Equation (1) described above.

(62) By executing the current amount predicting step S**7**, the amount of current that can be supplied by the fuel cell **14** is predicted.

(63) The current amount predicting step S**7** is executed after all of the first current amount calculating step S**1**, the hydrogen amount calculating step S**2**, and the remaining amount acquiring step S**3** are completed. In other words, the current amount predicting step S**7** may be started before the travel distance calculating step S**4**, the second current amount calculating step S**5**, and the remaining capacity acquiring step S**6**.

(64) In the electric power consumption calculating step S**8**, the electric power consumption calculating unit **42** calculates the electric power consumption EPC based on the execution results of the first current amount calculating step S**1**, the travel distance calculating step S**4**, and the second current amount calculating step S**5**. The electric power consumption calculating unit **42** can calculate the electric power consumption EPC by using, for example, Equation (2) described above.

(65) In the cruising distance calculating step S**9**, the cruising distance calculating unit **46** calculates the cruising distance CD based on the execution results of the remaining capacity acquiring step S**6**, the current amount predicting step S**7**, and the electric power consumption calculating step S**8**. The cruising distance calculating unit **46** can calculate the cruising distance CD using, for example, Equation (3) described above.

(66) In the display control step S**10**, the display control unit **48** controls the display unit **20** to

display the cruising distance CD calculated in the cruising distance calculating step S**9** on the display screen **201**.
(67) When the display control step S**10** is executed, the vehicle occupant in the vehicle **10** learns the cruising distance CD.
(68) [Modifications]
(69) Modifications of the above-described embodiment will be described below. However, description overlapping with the above embodiment will be omitted as much as possible in the following description. Constituent elements which have already been described in the above embodiment will be denoted with the same reference characters as those in the above embodiment unless otherwise indicated.
(70) (Modification 1)
(71) The vehicle **10** may further include a Fuel Cell Voltage Control Unit (FCVCU). The FCVCU includes, for example, a boost converter. The FCVCU is disposed between the fuel cell **14** and the rotating electric machine (inverter). The FCVCU adjusts the first current (voltage) and outputs the first current (voltage) to the drive source **18**. However, the first current adjusted by the FCVCU may be output to the battery **16**.
(72) In the vehicle **10** including the FCVCU, the first current amount calculating unit **30** may calculate the total amount of the first current after being adjusted by the FCVCU as the first total amount CA**1**. Therefore, the current sensor **50** may detect the current output from the FCVCU.
(73) (Modification 2)
(74) Based on the electric power consumption EPC and the predicted current amount CAP, the cruising distance calculating unit **46** may calculate a cruising distance CD (CD**2**) that the vehicle **10** can travel without supplying the remaining capacity CA**3** of the battery **16** to the rotating electric machine.
(75) By substituting 0 for CA**3** in Equation (3), following Equation (4) for calculating the cruising distance CD**2** is acquired. In Equation (4), CD**2** indicates the cruising distance CD**2**. The other characters conform to Equation (3).
$$CD2=CAP\times EPC \qquad (4)$$
(76) The display control unit **48** may control the display unit to display the cruising distance CD**2** on the display screen **201**.
(77) (Modification 3)
(78) In relation to Modification 2, the vehicle **10** need not necessarily include the battery **16**. In this case, the second current amount calculating unit **40**, the current sensor **56**, the remaining capacity acquisition unit **44**, and the remaining capacity sensor **58** are omitted from the configuration of the vehicle **10**.
(79) In addition, in the case of the present modification example, the method of calculating the electric power consumption EPC is different from that of the embodiment. Specifically, the electric power consumption EPC according to the present modification is a quotient obtained by dividing the travel distance TD by the first total amount CA**1**. Therefore, the electric power consumption calculating unit **42** according to the present modification calculates the electric power consumption EPC based on following Equation (5). In Equation (5), EPC**2** indicates the electric power consumption EPC according to the present modification.
(80) $$\mathrm{EPC2} = \frac{\mathrm{TD}}{\mathrm{CA1}} \qquad (5)$$
(Modification 4)
(81) Based on the electric power consumption EPC and the remaining capacity CA**3** of the battery **16**, the cruising distance calculating unit **46** may calculate a cruising distance CD (CD**3**) that the vehicle **10** can travel without power generation by the fuel cell **14**. For example, when the vehicle **10** travels in an EV mode, the vehicle **10** travels based on the remaining capacity CA**3**, without power generation of the fuel cell **14**.

(82) By substituting 0 for CAP in Equation (3), following Equation (6) for calculating the cruising distance CD**3** is acquired. In Equation (6), CD**3** indicates the cruising distance CD**3**.

$CD3=CA3\times EPC$ (6)

(83) The display control unit **48** may control the display unit **20** to display the cruising distance CD**3** on the display screen **201**.

(84) (Combination of Plural Modifications)

(85) The plurality of modifications described above may be appropriately combined as long as there is no contradiction in the combination.

Invention Obtained from Embodiment

(86) Hereinafter, invention that can be grasped from the above-described embodiment and modifications will be described.

(87) <First Aspect of Invention>

(88) According to a first aspect of the invention, the fuel cell vehicle (**10**) is equipped with the hydrogen tank (**12**) configured to store hydrogen, the fuel cell (**14**) configured to generate electric power using the hydrogen, and the drive source (**18**) configured to be driven using electric power generated by the fuel cell. The fuel cell vehicle includes the first current amount calculating unit (**30**) configured to calculate the first total amount (CA**1**) of output current that is output by the fuel cell during the period from the predetermined timing to the present time, the hydrogen amount calculating unit (**32**) configured to calculate the total amount (CHG) of hydrogen (hydrogen total amount) that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time, the remaining amount acquisition unit (**34**) configured to acquire the remaining amount (RHG) of hydrogen (hydrogen remaining amount) remaining in the hydrogen tank at the present time, and the current amount prediction unit (**36**) configured to predict the current amount (CAP) that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen.

(89) Consequently, it is possible to accurately predict the amount of current that can be supplied to the drive source by the fuel cell of the fuel cell vehicle.

(90) The first aspect of the invention may further include the travel distance calculating unit (**38**) configured to calculate the travel distance (TD) that the fuel cell vehicle has traveled from the predetermined timing to the present time, and the electric power consumption calculating unit (**42**) configured to calculate the electric power consumption (EPC) of the fuel cell vehicle based on the first total amount of the output current and the travel distance. Consequently, the electric power consumption reflecting the output current amount of the fuel cell is calculated.

(91) The first aspect of the invention may further include the battery (**16**) configured to supply electric power to the drive source, and the second current amount calculating unit (**40**) configured to calculate the second total amount (CA**2**) of the output current that is output by the battery during the period from the predetermined timing to the present time, wherein the electric power consumption calculating unit calculates the electric power consumption of the fuel cell vehicle based on the first total amount of the output current, the travel distance, and the second total amount. Consequently, the electric power consumption reflecting both the output current amount of the fuel cell and the output current amount of the battery is calculated.

(92) The first aspect of the fuel cell vehicle may further include the cruising distance calculating unit (**46**) configured to calculate the cruising distance (CD**2**) that the fuel cell vehicle is configured to travel only by the fuel cell, based on the electric power consumption of the fuel cell vehicle and the current amount predicted by the current amount prediction unit. Consequently, the cruising distance can be accurately predicted.

(93) The first aspect of the present invention may further include the remaining capacity acquisition unit (**44**) configured to acquire the remaining capacity (CA**3**) of the battery, and the cruising distance calculating unit (**46**) configured to calculate the cruising distance (CD**3**) that the fuel cell vehicle is configured to travel only by the battery, based on the electric power consumption of the

fuel cell vehicle and the remaining capacity of the battery. Consequently, the cruising distance can be accurately predicted.
(94) The cruising distance calculating unit may calculate the cruising distance (CD) that the fuel cell vehicle is configured to traveling by the battery and the fuel cell, based on the electric power consumption of the fuel cell vehicle, the current amount predicted by the current amount prediction unit, and the remaining capacity of the battery. Consequently, it is possible to accurately predict the cruising distance when the fuel cell vehicle travels in the EV mode, for example.
(95) The first aspect may further include the display unit (**20**) configured to display the cruising distance. Consequently, the vehicle occupant of the fuel cell vehicle learns the cruising distance.
(96) <Second Aspect of Invention>
(97) According to a second aspect of the invention, the computer predicts the amount of current that is configured to be supplied by the fuel cell (**14**) of the fuel cell vehicle (**10**) by the prediction method. The fuel cell vehicle including the hydrogen tank (**12**) configured to store hydrogen, the fuel cell configured to generate electric power using the hydrogen, and the drive source (**18**) configured to be driven using electric power generated by the fuel cell. The prediction method includes the first current amount calculating step (S**1**) of calculating the first total amount (CA**1**) of output current that is output by the fuel cell during the period from the predetermined timing to the present time, the hydrogen amount calculating step (S**2**) of calculating the total amount (CHG) of hydrogen (hydrogen total amount) that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time, the remaining amount acquiring step (S**3**) of acquiring the remaining amount (RHG) of hydrogen (hydrogen remaining amount) remaining in the hydrogen tank at the present time, and the current amount predicting step (S**7**) of predicting the current amount (CAP) that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen.
(98) Consequently, it is possible to accurately predict the amount of current that can be supplied to the drive source by the fuel cell of the fuel cell vehicle.
(99) The present invention is not limited to the above disclosure, and various modifications are possible without departing from the essence and gist of the present invention.

Claims

1. A fuel cell vehicle that includes a hydrogen tank configured to store hydrogen, a fuel cell configured to generate electric power using the hydrogen, a battery, and a drive source configured to be driven using the electric power generated by the fuel cell or supplied from the battery, the fuel cell vehicle comprising one or more processors that execute computer-executable instructions stored in a memory, wherein the one or more processors execute the computer-executable instructions to cause the fuel cell vehicle to: calculate a first total amount of output current that is output by the fuel cell during a period from a predetermined timing to present time; calculate a total amount of hydrogen that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time; acquire a remaining amount of hydrogen remaining in the hydrogen tank at the present time; predict, as a predicted current amount, a current amount that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen; calculate a total travel distance that the fuel cell vehicle has traveled during the period from the predetermined timing to the present time; calculate an electric power consumption of the fuel cell vehicle based on the first total amount of the output current and the total travel distance, the electric power consumption indicating a travel distance of the fuel cell vehicle per total amount of current supplied to the drive source; acquire a remaining capacity of the battery; and calculate a cruising distance that the fuel cell vehicle is configured to travel, based on the predicted current amount, the electric power consumption of the fuel cell vehicle, and the remaining capacity of the battery, wherein the one or

more processors cause the fuel cell vehicle to calculate the cruising distance as a first cruising distance that the fuel cell vehicle is configured to travel by the current supplied to the drive source only from the fuel cell, based on the predicted current amount and the electric power consumption of the fuel cell vehicle.

2. The fuel cell vehicle according to claim 1, wherein the one or more processors cause the fuel cell vehicle to: calculate a second total amount of output current that is output by the battery during the period from the predetermined timing to the present time, and calculate the electric power consumption of the fuel cell vehicle based on the first total amount of the output current, the total travel distance, and the second total amount.

3. The fuel cell vehicle according to claim 1, wherein the one or more processors cause the fuel cell vehicle to: calculate the cruising distance as a second cruising distance that the fuel cell vehicle is configured to travel only by the battery, based on the electric power consumption of the fuel cell vehicle and the remaining capacity of the battery.

4. The fuel cell vehicle according to claim 1, wherein the one or more processors cause the fuel cell vehicle to calculate a the cruising distance as a third cruising distance that the fuel cell vehicle is configured to travel by the battery and the fuel cell, based on the electric power consumption of the fuel cell vehicle, the predicted current amount, and the remaining capacity of the battery.

5. The fuel cell vehicle according to claim 1, wherein the one or more processors cause the fuel cell vehicle to display the cruising distance.

6. A prediction method for predicting an amount of current that is configured to be supplied by a fuel cell of a fuel cell vehicle, the fuel cell vehicle including a hydrogen tank configured to store hydrogen, a battery, and a drive source configured to be driven using the electric power generated by the fuel cell or supplied from the battery, the prediction method comprising: calculating a first total amount of output current that is output by the fuel cell during a period from a predetermined timing to present time; calculating a total amount of hydrogen that is output from the hydrogen tank to the fuel cell during the period from the predetermined timing to the present time; acquiring a remaining amount of hydrogen remaining in the hydrogen tank at the present time; and predicting, as a predicted current amount, a current amount that is configured to be supplied by the fuel cell based on the first total amount of the output current, the total amount of hydrogen, and the remaining amount of hydrogen; calculating a total travel distance that the fuel cell vehicle has traveled during the period from the predetermined timing to the present time; calculating an electric power consumption of the fuel cell vehicle based on the first total amount of the output current and the total travel distance, the electric power consumption indicating a travel distance of the fuel cell vehicle per total amount of current supplied to the drive source; acquiring a remaining capacity of the battery; and calculating a cruising distance that the fuel cell vehicle is configured to travel, based on the predicted current amount, the electric power consumption of the fuel cell vehicle, and the remaining capacity of the battery, wherein in calculating the cruising distance, the cruising distance is calculated as a first cruising distance that the fuel cell vehicle is configured to travel by the current supplied to the drive source only from the fuel cell, based on the predicted current amount and the electric power consumption of the fuel cell vehicle.